VAE notes: β-VAE, conditional VAE and latent space interpolation

Study notes for NPTEL Fundamentals of Generative AI and LLMs, Week 4, Lectures 26 to 28

What the three lectures cover

- **Lecture 26:** entangled and disentangled latent spaces, and β-VAE as a way to encourage disentanglement.
- **Lecture 27:** the conditional VAE (cVAE), which controls what gets generated by feeding in a label.
- **Lecture 28:** latent space interpolation, which moves smoothly between two images through the latent space.

Each lecture builds on the VAE basics below, so start there.

VAE recap

- The **encoder** maps an input x to a distribution over the latent space: a mean $\mu_\varphi(x)$ and a variance $\sigma_\varphi^2(x)$. Written as $q_\varphi(z \mid x) = N(\mu_\varphi(x), \sigma_\varphi^2(x))$.
- A latent vector is sampled with the **reparameterization trick**: $z = \mu_\varphi(x) + \sigma_\varphi(x) \odot \varepsilon$, where $\varepsilon \sim N(0, I)$. The symbol ⊙ means element-wise multiplication.
- The **decoder** $p_\theta(x \mid z)$ turns z back into data.
- A standard VAE therefore learns x → z → $\hat{x}$.
- The loss has two parts: $L(x) = -E[\log p_\theta(x \mid z)] + D_{KL}(q_\varphi(z \mid x) \| p(z))$. The first term is the **reconstruction loss** and the second is the **KL loss**. The prior is $p(z) = N(0, I)$.

Why the trick exists (standard background, not on the slides): writing z as a mean plus scaled noise lets gradients flow back through the sampling step to the encoder.

Lecture 26: entanglement, disentanglement and β-VAE

Entangled latent space

Take a 28 × 28 image. It has 784 pixel values, but the important variations are far fewer, for example thickness, slant, rotation and position.

The encoder compresses the image into a small latent vector, say `z = [z1, z2, z3, z4]`. Every hidden unit is a combination of all 784 pixels, so each latent variable ends up holding a mix of thickness, slant, rotation and position.

> In an **entangled** latent representation, changing one latent variable may change several properties together.

Disentangled latent space

A **disentangled** latent space aims to learn a structured representation where each latent dimension mainly captures one meaningful factor of variation.

Face example with latent size 5:

Latent variable	Mainly controls
z1	hair
z2	lighting / overexposure
z3	skin tone
z4	smile
z5	pose

Change only z1 and mainly the hair changes, while lighting, skin tone, smile and pose stay mostly the same. This makes the latent space **interpretable** and **controllable**, because manipulating one latent variable gives a controlled change in the image.

β-VAE: encouraging disentanglement

The standard VAE loss has the KL term with weight 1. The KL term is still there, but the model may accept mixed information spread across many latent variables if that helps reconstruction. So even after training, the latents can stay entangled: **a standard VAE**

does not guarantee disentanglement.

β-VAE adds a weight β to the KL term:

`L_β-VAE(x) = −E[log pθ(x | z)] + β · D_KL(qφ(z | x) || p(z))`, with **β > 1**.

The two terms pull in opposite directions:

- **Reconstruction loss:** store enough information in z that the decoder can rebuild x ($\hat{x} \approx x$).
- **KL loss:** keep $q_\varphi(z \mid x)$ close to the standard normal prior, so information isn't stored in an arbitrary way. This pushes $\mu(x) \approx 0$ and $\sigma^2(x) \approx 1$.

The model is balancing **good reconstruction against a regular latent space**. A larger β puts more pressure on the KL term, which stops the model from freely storing mixed information in all the latent variables.

Two caveats from the slides:

- β-VAE encourages disentanglement through stronger regularization, but it **does not guarantee perfect disentanglement**.
- If β is too large, **reconstruction quality may decrease**, because the model focuses too much on staying close to N(0, I) and may not store enough detail.

Worked KL examples

For one latent dimension, `D_KL = ½(μ² + σ² − log σ² − 1)`. With $\sigma^2 = 1$ this becomes `½μ²`. For two dimensions: `D_KL = ½(μ1² + μ2²)`. All three examples assume $\sigma_1^2 = \sigma_2^2 = 1$.

Case	μ1	μ2	Calculation	D_KL
Disentangled example	0.4	0.2	½(0.16 + 0.04)	**0.10**
Entangled example 1	0.8	0.6	½(0.64 + 0.36)	**0.50**
Entangled example 2	1.5	0.5	½(2.25 + 0.25)	**1.25**

Reading the result: a small KL means the distribution is close to the prior (μ near [0, 0]) and more regularized. A larger KL means it is farther from the prior and the encoder is using more latent capacity to store information.

When there are more factors than latent variables

Suppose the image has 6 important factors (hair, lighting, skin tone, smile, pose, background) but the latent size is only 3. There are two cases.

Case 1: entanglement happens. One latent variable captures a mixture of factors. More information is stored, so reconstruction loss may be lower, but the encoder moves farther from the prior, so KL may increase. The β value is usually low or not strong enough to enforce disentanglement.

Case 2: disentanglement is encouraged. Each latent variable is pushed to capture one factor, so the model captures only a few dominant factors and the rest are ignored or weakly represented. The decoder gets incomplete information, so reconstruction loss increases. With $\beta > 1$ the KL term gets more weight, so KL may decrease.

Three possible fixes

1. **Increase the latent size**, for example from 3 to 6 variables. More capacity means more information captured, better reconstruction and lower reconstruction loss.
2. **Tune β.** A very high β means strong KL pressure, less information in z and higher reconstruction loss, so choose a moderate β to balance reconstruction and disentanglement.
3. **β annealing.** Don't use a high β from the start. Begin small and increase it gradually (for example 1 → 2 → 4), or start from 0 or a very small value so the model learns reconstruction first, then raise β so the KL term becomes active and encourages disentanglement.

Lecture 27: conditional VAE (cVAE)

The problem it solves

In a standard VAE, generation is not explicitly controlled. If you want to generate a specific kind of output, such as the digit 5 or a cat image, you need extra guidance.

A **conditional VAE** is a VAE that gets an extra input called a **condition y**, such as a class label or attribute. A standard VAE learns x → z → x̂. A cVAE learns **(x, y) → z → x̂**, which gives **controlled generation**.

How the label is used

- The **encoder** uses both x and y: `qφ(z | x, y)`.

- The **decoder** uses both z and y: `pθ(x | z, y)`.
- The label is joined to the input by concatenation (the ⊕ in the diagram), and to z before the decoder.
- Sampling stays the same: `z = μφ(x, y) + σφ(x, y) ⊙ ε`.

Loss function

Standard VAE: `L(x) = −E_q(z|x)[log pθ(x | z)] + D_KL(qφ(z | x) ‖ p(z))`

cVAE: `L_cVAE(x) = −E_q(z|x,y)[log pθ(x | z, y)] + D_KL(qφ(z | x, y) ‖ p(z))`

The only change is that y is added inside the encoder and decoder terms. The prior p(z) stays the same.

Numerical example

Item	Standard VAE	cVAE
Encoder input	x = [5.1, 3.5, 1.4, 0.2]	[x, y] = [5.1, 3.5, 1.4, 0.2, 1, 0] with y = [1, 0]
Encoder hidden activation	[1.07, 1.93, 0.87]	[1.27, 1.83, 1.17]
Decoder input	z	[z, y] = [0.917, 0.153, 1, 0] with z = [0.917, 0.153]
Reconstructed output	[0.103, −0.028, 0.206, 0.243]	[0.147, −0.047, 0.280, 0.337]

Key ideas from the example:

- The label contributes additional weighted information to the encoder's hidden layer, which changes the hidden activation.
- The decoder is told to reconstruct x using the latent information z, **but conditioned on the class label y = [1, 0]**.
- The label changes the decoder's hidden representation and therefore the reconstructed output.

Lecture 28: latent space interpolation

The idea

Take two images, a start image A (digit 3) and an end image B (digit 8). Encode both to get two points in the latent space, move between them in steps, and decode each intermediate point. The decoder output changes smoothly from one image to the other.

Step 1: get the two latent points

The encoder gives μ and σ² for each image, and z is sampled with $z = \mu + \sigma \odot \varepsilon$.

Image	μ	z
A (digit 3)	[0.043, 0.551]	z_A = [0.764, 0.318]
B (digit 8)	[−1.101, −0.993]	z_B = [−0.200, 1.100]

Each z is one point in the 2-dimensional latent space.

Step 2: interpolate

$z_\alpha = (1 - \alpha) \cdot z_A + \alpha \cdot z_B$

α controls the movement from z_A to z_B. At **α = 0** you get exactly z_A (image A) and at **α = 1** you get exactly z_B (image B). Values in between give points between A and B.
Because the latent space is 2-dimensional, it is applied per component:

$z_\alpha = [(1 - \alpha)\ z_{A1} + \alpha\ z_{B1},\ (1 - \alpha)\ z_{A2} + \alpha\ z_{B2}]$

Worked example for α = 0.25:

- First component: 0.75(0.764) + 0.25(−0.200) = 0.573 − 0.050 = 0.523
- Second component: 0.75(0.318) + 0.25(1.100) = 0.2385 + 0.275 = 0.5135
- So z_0.25 = [0.523, 0.514]

All five points:

α	Interpolated latent vector z_α
0	[0.764, 0.318]
0.25	[0.523, 0.514]

α	Interpolated latent vector z_α
0.5	[0.282, 0.709]
0.75	[0.041, 0.9045]
1	[−0.200, 1.100]

Step 3: decode

Each interpolated vector goes through the decoder: z_A → Decoder → x̂_A, z_0.25 → Decoder → x̂_0.25, and so on up to z_B → Decoder → x̂_B.

The generated images change like this:

Digit 3 → 3-like intermediate image → mixed 3/8 image → 8-like intermediate image → Digit 8

This is called latent space interpolation because we are moving smoothly between two latent points in the learned latent space. The lecture also shows the same idea on 3D chair shapes, morphing from one chair to another.

One practical note (background, not on the slides): the lecture interpolates between sampled z points. Many implementations interpolate between the means μ instead, so the result is the same each time.

The three models side by side

	Standard VAE	β-VAE	cVAE
Learns	x → z → x̂	x → z → x̂	(x, y) → z → x̂
What changes	Nothing, it is the baseline	KL term multiplied by β > 1	Label y fed to encoder and decoder
Goal	Compressed latent representation	Encourage disentangled latents	Controlled generation
Weakness	Doesn't guarantee disentanglement	Too large β hurts reconstruction	Needs labels

Formula sheet

- Reparameterization: $z = \mu_\varphi(x) + \sigma_\varphi(x) \odot \varepsilon$, $\varepsilon \sim N(0, I)$
- VAE loss: $-E[\log p_\theta(x \mid z)] + D_{KL}(q_\varphi(z \mid x) \,\|\, p(z))$
- β-VAE loss: $-E[\log p_\theta(x \mid z)] + \beta \cdot D_{KL}(q_\varphi(z \mid x) \,\|\, p(z))$, $\beta > 1$
- KL per dimension: $\frac{1}{2}(\mu^2 + \sigma^2 - \log \sigma^2 - 1)$, which is $\frac{1}{2}\mu^2$ when $\sigma^2 = 1$
- cVAE loss: $-E[\log p_\theta(x \mid z, y)] + D_{KL}(q_\varphi(z \mid x, y) \,\|\, p(z))$
- Interpolation: $z_\alpha = (1 - \alpha)\, z_A + \alpha\, z_B$

Likely exam questions

What is an entangled latent representation? One where each latent variable holds a mix of several factors, so changing one variable changes multiple properties together.

Does a standard VAE guarantee disentanglement? No. With KL weight 1 the model can still accept mixed information across latents if that helps reconstruction.

What does β do in β-VAE, and what is the trade-off? It weights the KL term ($\beta > 1$), pushing the encoder distribution toward N(0, I) and encouraging disentanglement. If β is too large, reconstruction quality drops.

Compute the KL for $\mu = [0.6, 0.8]$ with $\sigma^2 = 1$. $\frac{1}{2}(0.36 + 0.64) = 0.50$.

How does a cVAE differ from a VAE? A cVAE adds a condition y, such as a class label. The encoder becomes $q_\varphi(z \mid x, y)$, the decoder becomes $p_\theta(x \mid z, y)$, and the loss uses both, which allows controlled generation.

What is the decoder input in a cVAE with $z = [0.917, 0.153]$ and $y = [1, 0]$? [0.917, 0.153, 1, 0].

Compute $z_{0.5}$ for $z_A = [0.764, 0.318]$ and $z_B = [-0.200, 1.100]$. $0.5 \times z_A + 0.5 \times z_B$ = [0.282, 0.709].

Give three ways to handle too many factors and too few latent variables. Increase the latent size, tune β to a moderate value, or use β annealing.

Why is interpolation done in latent space and not on pixels? Blending the pixels of a 3 and an 8 would just overlay two images. Moving between latent points and decoding gives intermediate images that look like real digits, which is the point of the lecture.

Common mix-ups

- β-VAE uses **β > 1**, not β < 1.
- A smaller KL means closer to the prior and more regularized. It doesn't mean the model is better, since reconstruction can suffer.
- In a cVAE the label goes into **both** the encoder and the decoder.
- At α = 0 you get z_A, and at α = 1 you get z_B, not the other way round.
- β annealing means starting with a small β and increasing it, not the reverse.

Part 2: deeper explanations

Part 1 above is the summary. This part explains the ideas behind it, so you can answer questions that aren't copied straight from a slide. Anything marked *background* is standard material that the slides assume rather than state.

From autoencoder to VAE (background)

A plain autoencoder maps each input to one fixed point in the latent space and decodes from that point. Nothing forces the space between those points to decode into anything sensible. A VAE instead maps each input to a distribution (μ, σ²) and samples from it, and the KL term keeps all those distributions close to N(0, I). The result is a latent space where nearby points decode to similar outputs, and where sampling from N(0, I) generates new data. Everything in this week relies on that.

What each part of the loss does

- **Reconstruction term** `-E[log pθ(x | z)]`: measures how badly the decoder rebuilds x from a sampled z. Lower is better.
- **KL term** `D_KL(qφ(z | x) || p(z))`: measures how far the encoder's distribution is from N(0, I).
- **The tension:** to reconstruct well, z must carry a lot of information about x, so different inputs need different μ and small σ. The KL term wants every input to look like the prior (μ = 0, σ² = 1). The trained model is a compromise between the two.

Reading the KL formula

For each latent dimension j:

$$D_{KL,j} = \tfrac{1}{2}\left(\mu_j^2 + \sigma_j^2 - \ln \sigma_j^2 - 1\right)$$

The total KL is the sum over dimensions.

- `μj²` is the penalty for the mean being away from 0. It grows with the square, so doubling μ quadruples this part.
- `σj² − ln σj² − 1` is exactly 0 when σj² = 1 and positive otherwise. It penalizes variance that is bigger **or** smaller than 1.
- A latent dimension that carries no information sits at μ = 0, σ² = 1 and costs nothing.
- The slides assume σ² = 1 so only the μ term is left. That is why the lecture arithmetic is so short.

Per-dimension KL when μ = 0:

σ²	KL for that dimension
0.5	0.097
1	0
2	0.153
4	0.807

Entanglement, step by step

The slide's network for a 28 × 28 image: x (784 values) → hidden layer h (8 units) → μ and σ² (4 values each) → z (4 values).

- Each hidden unit h1, ..., h8 is a weighted combination of **all** 784 pixels.
- Each latent value is a weighted combination of **all** the hidden units.
- Nothing in this architecture assigns a latent to a factor like thickness or slant.

So unless something encourages it, the factors get spread across the latents. Entanglement is the default outcome, and disentanglement is what you have to push the model toward.

How you would check disentanglement: latent traversal (background)

Encode an image, then vary **one** latent z_i over a range while holding the others fixed, and decode each version. If one visible property changes (say only the smile), that dimension is disentangled. If several properties change together, it is entangled. This is the same idea as interpolation, but along a single axis.

Why $\beta > 1$ helps, and what it costs

The slides' argument: a larger KL weight pulls $q_\phi(z \mid x)$ harder toward the prior, which restricts the model from freely storing mixed information in all the latents.

A common way to say it (background): a larger β limits how much information z can carry. The model then spends its limited capacity on the factors that matter most for reconstruction, tends to give each one its own dimension, and that lines up with independent factors.

The cost is that less information in z means worse reconstruction. So β is always a balance between reconstruction and disentanglement.

How the cases compare:

Situation	Reconstruction loss	KL	What the latents look like
Low β, few latents (entanglement)	Lower	Higher	Mixed factors in each latent
High β, few latents (disentanglement enforced)	Higher	Lower	Only a few dominant factors captured
Larger latent size, moderate β	Lower	Moderate	More factors can have their own latent

The three fixes, and why each works

1. **Increase latent size.** More capacity means more factors can be captured separately, so reconstruction improves.
2. **Tune β to a moderate value.** Too high hurts reconstruction, too low doesn't enforce disentanglement.
3. **β annealing.** Let the model learn reconstruction first with a small β (the slides give both 1 → 2 → 4 and starting from 0 or a very small value), then raise β gradually so the KL term becomes active and encourages disentanglement. Either way the rule is the same: start small, increase slowly.

Conditional VAE in detail

- **Training data** is pairs (x, y), such as an image and its class.
- The **encoder** gets the concatenation [x, y]. The **decoder** gets [z, y].

- The label is one-hot for classes: with two classes, y = [1, 0] or [0, 1]. Its length equals the number of classes.
- Because y is supplied to the decoder separately, z doesn't have to spend capacity encoding the class, and can focus on the other variation, such as style (background).
- **Generation** (background): there is no x at generation time. Sample z from N(0, I), choose the label you want, such as the one-hot for the digit 5, and feed [z, y] to the decoder. Different z with the same y give different versions of the same class.
- The loss differs from the VAE loss only by conditioning on y. The prior p(z) is unchanged on the slides.

Reading the cVAE numerical example

The first layer of the encoder computes a weighted sum of its inputs. With the label attached, that sum has an extra piece from y. With y = [1, 0], that extra piece is just the weights attached to the first label unit.

On the slide the hidden activation moves from [1.07, 1.93, 0.87] to [1.27, 1.83, 1.17]. The change is **[+0.20, −0.10, +0.30]**, and that is the contribution of the label. The decoder output moves from [0.103, −0.028, 0.206, 0.243] to [0.147, −0.047, 0.280, 0.337], a change of [+0.044, −0.019, +0.074, +0.094]. This reading is an inference from the architecture, not something the slide states, but it explains the slide's key line: the label contributes additional weighted information to the hidden layer.

Latent space interpolation in detail

The procedure:

1. Encode image A and image B to get z_A and z_B.
2. Compute z_α for several values of α.
3. Decode every z_α.

The geometry. z_α lies on the straight line from z_A to z_B. α is the fraction of the way along that line. An equivalent form is:

```
z_α = z_A + α (z_B − z_A)
```

With the slide numbers, z_B − z_A = [−0.964, 0.782]. For α = 0.25: z_A + 0.25 × [−0.964, 0.782] = [0.764 − 0.241, 0.318 + 0.1955] = [0.523, 0.5135]. This matches the lecture's answer.

Other points worth knowing:

- The two weights, $(1 - \alpha)$ and α, always add up to 1, so z_α is a weighted average of the two endpoints.
- Equally spaced α values give equally spaced points in latent space. That doesn't always mean equal visual change in the images.
- α outside $[0, 1]$ is extrapolation: it goes past z_A ($\alpha < 0$) or past z_B ($\alpha > 1$).
- Interpolation must happen in latent space. Averaging the pixels of a 3 and an 8 gives a ghosted overlay, not an intermediate digit. A regularized latent space is what makes the in-between points decode to plausible digits (background).
- The lecture interpolates between sampled z points. Many implementations use the means μ instead, so the output doesn't change from run to run (background).
- Uses (background): morphing between images, checking that the latent space is smooth, and exploring what the model has learned.

How the three ideas fit together

- **β-VAE** shapes the latent space so its dimensions are meaningful.
- **cVAE** lets you steer what gets generated by supplying a label.
- **Interpolation** lets you explore and demonstrate the latent space.

They are separate tools and can be combined, for example a cVAE trained with $\beta > 1$.

Part 3: numericals with full solutions

Use the natural log (ln) wherever the slides write log. When $\sigma^2 = 1$ the log term vanishes, which is why the lecture examples use $\sigma^2 = 1$. Try each problem on paper before reading the solution.

A. Reparameterization

N1. An encoder gives $\mu = [0.2, -0.5]$, $\sigma = [0.5, 2]$, and the sampled noise is $\varepsilon = [1, -0.5]$. Find z.

Solution: $z = \mu + \sigma \odot \varepsilon$, element by element. $z1 = 0.2 + 0.5 \times 1 = 0.7$. $z2 = -0.5 + 2 \times (-0.5) = -1.5$.

Answer: z = [0.7, −1.5].

N2. $\mu = [1, 0, -1]$, $\sigma^2 = [4, 1, 0.25]$, $\varepsilon = [0.5, -1, 2]$. Find z.

Solution: the encoder gives variances, so take the square root first: σ = [2, 1, 0.5]. z1 = 1 + 2 × 0.5 = 2. z2 = 0 + 1 × (−1) = −1. z3 = −1 + 0.5 × 2 = 0.

Answer: z = [2, −1, 0].

N3. What is z when ε = 0? Give z for μ = [0.043, 0.551].

Solution: the noise term vanishes, so z = μ.

Answer: z = [0.043, 0.551]. This is why some implementations use μ directly when they want a deterministic result.

N4. For image A the lecture has μ_A = [0.043, 0.551] and z_A = [0.764, 0.318]. Assuming ε = [1.2, −0.4], find σ and σ².

Solution: z − μ = σ ⊙ ε, so σ = (z − μ) / ε. z − μ = [0.721, −0.233]. σ1 = 0.721 / 1.2 = 0.601. σ2 = −0.233 / −0.4 = 0.5825. Squaring gives σ² = [0.361, 0.339].

Answer: σ ≈ [0.601, 0.583], σ² ≈ [0.361, 0.339]. (The ε values are assumed for practice; the slide doesn't give them.)

B. KL divergence

N5. μ = [1, −1, 2], σ² = 1 in every dimension. Find D_KL.

Solution: with σ² = 1, D_KL = ½ Σ μj² = ½ (1 + 1 + 4) = ½ × 6.

Answer: 3.

N6. μ = [0.3, 0.4], σ² = [1, 1]. Find D_KL.

Solution: ½ (0.09 + 0.16) = ½ × 0.25.

Answer: 0.125.

N7. One dimension with μ = 0 and σ² = 2. Find D_KL.

Solution: ½ (0 + 2 − ln 2 − 1) = ½ (1 − 0.6931) = ½ × 0.3069.

Answer: 0.153.

N8. One dimension with μ = 0 and σ² = 0.5. Find D_KL.

Solution: ½ (0 + 0.5 − ln 0.5 − 1) = ½ (0.5 + 0.6931 − 1) = ½ × 0.1931.

Answer: 0.097. Note that a variance below 1 is also penalized.

N9. One dimension with μ = 1 and σ² = 4. Find D_KL.

Solution: ½ (1 + 4 − ln 4 − 1) = ½ (4 − 1.3863) = ½ × 2.6137.

Answer: 1.307.

N10. Two dimensions: $\mu = [0.5, -0.5]$, $\sigma^2 = [1, 0.25]$. Find D_{KL}.

Solution: dimension 1: ½ (0.25 + 1 − 0 − 1) = 0.125. Dimension 2: ½ (0.25 + 0.25 − ln 0.25 − 1) = ½ (0.5 + 1.3863 − 1) = 0.4431. Total = 0.125 + 0.4431.

Answer: 0.568.

N11. Encoder P gives $\mu = [0.2, 0.1]$ and encoder Q gives $\mu = [1.2, 0.9]$, both with $\sigma^2 = 1$. Compare their KL values and say which is more regularized.

Solution: KL_P = ½ (0.04 + 0.01) = 0.025. KL_Q = ½ (1.44 + 0.81) = 1.125. Q's KL is 45 times larger.

Answer: P is far more regularized and closer to the prior. Q uses more latent capacity to store information.

N12. With $\sigma^2 = 1$ in one dimension, what μ gives $D_{KL} = 2$?

Solution: ½ $\mu^2 = 2$, so $\mu^2 = 4$.

Answer: $\mu = \pm 2$.

N13. The disentangled example has $\mu = [0.4, 0.2]$ and KL 0.10. What is the KL if both means are doubled to $[0.8, 0.4]$?

Solution: ½ (0.64 + 0.16) = 0.40.

Answer: 0.40. Doubling μ multiplied the KL by 4, because KL grows with μ^2.

C. β-weighted loss

N14. A model has reconstruction loss 2.0 and KL 0.5. Find the total loss for $\beta = 1$, 4 and 10.

Solution: total = reconstruction + β × KL. β = 1: 2.0 + 0.5 = 2.5. β = 4: 2.0 + 2.0 = 4.0. β = 10: 2.0 + 5.0 = 7.0.

Answer: 2.5, 4.0, 7.0.

N15. Three models are trained with different β:

β	Reconstruction loss	KL
1	1.2	3.0
4	1.8	1.4

β	Reconstruction loss	KL
10	3.0	0.6

Find each total loss. Can you say the β = 1 model is best because its total is lowest?

Solution: β = 1: 1.2 + 3.0 = 4.2. β = 4: 1.8 + 4 × 1.4 = 7.4. β = 10: 3.0 + 10 × 0.6 = 9.0.

Answer: 4.2, 7.4, 9.0. No. The totals use different objectives, so they can't be compared. Compare the reconstruction and KL columns instead: as β rises, KL falls (more regularized) and reconstruction gets worse. That is exactly the trade-off.

N16. β is annealed linearly as β = 0.5 × epoch. At epoch 3 the reconstruction loss is 1.0 and the KL is 2.0. Find the total loss at epoch 3, and at epoch 8 if the losses stay the same.

Solution: epoch 3: β = 1.5, so total = 1.0 + 1.5 × 2.0 = 4.0. Epoch 8: β = 4, so total = 1.0 + 4 × 2.0 = 9.0.

Answer: 4.0 and 9.0. (Early on β is small, so the KL term hardly matters and the model learns reconstruction first.)

N17. A cVAE has reconstruction term 0.9 and KL term 0.3, with the standard weighting. Find the loss.

Solution: L_cVAE = 0.9 + 1 × 0.3.

Answer: 1.2.

N18. Four β-VAE runs:

β	Reconstruction loss	KL
1	0.5	5.0
2	0.6	3.5
4	0.8	2.0
8	1.6	1.2

If you want better disentanglement without losing too much reconstruction quality, which β looks most sensible?

Solution: from β = 1 to β = 4 the KL drops from 5.0 to 2.0 while reconstruction worsens only from 0.5 to 0.8. Going on to β = 8 doubles the reconstruction loss (0.8 to 1.6) for a small KL gain (2.0 to 1.2).

Answer: β = 4 is the balanced choice. This is a judgement question: the point is the trade-off, and the next step (β = 8) shows the reconstruction cost of too large a β.

D. Interpolation

Use z_A = [0.764, 0.318] and z_B = [−0.200, 1.100] unless a question says otherwise.

N19. Find z_α for α = 0.1.

Solution: 0.9 × [0.764, 0.318] + 0.1 × [−0.2, 1.1] = [0.6876 − 0.02, 0.2862 + 0.11].

Answer: [0.6676, 0.3962].

N20. Find z_α for α = 0.6.

Solution: 0.4 × [0.764, 0.318] + 0.6 × [−0.2, 1.1] = [0.3056 − 0.12, 0.1272 + 0.66].

Answer: [0.1856, 0.7872].

N21. In a 3-dimensional latent space z_A = [1, 2, 3] and z_B = [3, 0, −1]. Find z_α for α = 0.25, 0.5 and 0.75.

Solution: α = 0.25: 0.75 × [1, 2, 3] + 0.25 × [3, 0, −1] = [0.75 + 0.75, 1.5 + 0, 2.25 − 0.25]. α = 0.5: the average of the two points. α = 0.75: 0.25 × [1, 2, 3] + 0.75 × [3, 0, −1] = [0.25 + 2.25, 0.5 + 0, 0.75 − 0.75].

Answer: [1.5, 1.5, 2.0], [2, 1, 1], [2.5, 0.5, 0].

N22. For which α does the first component of z_α equal 0? What is z_α there?

Solution: (1 − α)(0.764) + α(−0.2) = 0, so 0.764 − 0.964α = 0, giving α = 0.764 / 0.964 = 0.7925. Second component: 0.2075 × 0.318 + 0.7925 × 1.1 = 0.0660 + 0.8718 = 0.9378.

Answer: α ≈ 0.7925, z_α ≈ [0, 0.938].

N23. You want 6 equally spaced images between z_A = [0, 4] and z_B = [10, −1]. Give the α values and the latent vectors.

Solution: 6 points means 5 steps, so α = 0, 0.2, 0.4, 0.6, 0.8, 1. The formula simplifies to z_α = [10α, 4 − 5α].

α	z_α
0	[0, 4]
0.2	[2, 3]
0.4	[4, 2]

α	z_α
0.6	[6, 1]
0.8	[8, 0]
1	[10, −1]

N24. Find the straight-line distance between z_A and z_B, and the distance between consecutive points when α goes 0, 0.25, 0.5, 0.75, 1.

Solution: z_B − z_A = [−0.964, 0.782]. Distance = $\sqrt{(0.964^2 + 0.782^2)} = \sqrt{(0.9293 + 0.6115)} = \sqrt{1.5408} = 1.2413$. There are 4 equal steps, so each is 1.2413 / 4 = 0.3103.

Answer: 1.241 in total, 0.310 per step.

N25. Check the lecture's α = 0.75 answer using the form z_A + α (z_B − z_A).

Solution: 0.75 × [−0.964, 0.782] = [−0.723, 0.5865]. Add z_A: [0.764 − 0.723, 0.318 + 0.5865].

Answer: [0.041, 0.9045], which matches the slide.

N26. Find z_α for α = 1.25. What does it mean?

Solution: (1 − 1.25) z_A + 1.25 z_B = −0.25 × [0.764, 0.318] + 1.25 × [−0.2, 1.1] = [−0.191 − 0.25, −0.0795 + 1.375].

Answer: [−0.441, 1.2955]. Since α > 1, this is extrapolation: a point beyond z_B on the same line.

E. Conditional VAE

N27. An MNIST cVAE uses 28 × 28 images, a one-hot label over 10 digits, and a latent size of 20. Find the encoder input size and the decoder input size.

Solution: encoder input = 784 + 10 = 794. Decoder input = 20 + 10 = 30.

Answer: 794 and 30.

N28. x = [5.1, 3.5, 1.4, 0.2] belongs to the second of 3 classes. Write y and the cVAE encoder input.

Solution: one-hot for the second class is [0, 1, 0]. Concatenate it to x.

Answer: y = [0, 1, 0], [x, y] = [5.1, 3.5, 1.4, 0.2, 0, 1, 0] (7 values).

N29. The first encoder layer has 256 units. How many extra weights does the cVAE need compared with a plain VAE for the 784-pixel input with a 10-class label? Then do the same for a decoder first layer with a latent size of 20.

Solution: encoder: (794 − 784) × 256 = 2,560 extra weights. Decoder: the input goes from 20 to 30, so (30 − 20) × 256 = 2,560 extra weights.

Answer: 2,560 in each. Conditioning is cheap.

N30. Using the slide values, find the change caused by the label in (a) the encoder hidden activation and (b) the reconstructed output.

Solution: (a) [1.27, 1.83, 1.17] − [1.07, 1.93, 0.87] = [0.20, −0.10, 0.30]. (b) [0.147, −0.047, 0.280, 0.337] − [0.103, −0.028, 0.206, 0.243] = [0.044, −0.019, 0.074, 0.094].

Answer: [0.20, −0.10, 0.30] and [0.044, −0.019, 0.074, 0.094].

F. Counting for the entanglement network

N31. The lecture's network is 784 inputs → 8 hidden units → μ (4 values) and σ^2 (4 values). Count the encoder's weights and biases.

Solution: input to hidden: 784 × 8 = 6,272 weights, plus 8 biases. Hidden to μ: 8 × 4 = 32 weights, plus 4 biases. Hidden to σ^2: 8 × 4 = 32 weights, plus 4 biases. Weights: 6,272 + 32 + 32 = 6,336. Biases: 8 + 4 + 4 = 16.

Answer: 6,336 weights, 16 biases, 6,352 parameters in total.

N32. By what factor does this encoder compress the image into z?

Solution: 784 values down to 4 latent values: 784 / 4 = 196.

Answer: 196 times fewer numbers. That heavy compression is why each latent ends up carrying many properties unless something pushes it not to.

Part 4: multiple choice questions

Sixty questions in three blocks, one per lecture, plus a mixed block. Cover the answers and try each one first.

Lecture 26: entanglement, disentanglement and β-VAE

Q1. In a standard VAE the encoder outputs:

- A) one fixed latent point z

- B) the reconstructed image $\hat{x}$
- C) a mean μ and variance σ^2 describing a distribution over z
- D) a class label

Answer: C. The encoder gives the parameters of $q_\phi(z \mid x)$, and z is then sampled from it.

Q2. The reparameterization trick samples z as:

- A) $z = \mu \times \sigma$
- B) $z = \mu + \sigma \odot \varepsilon$ with $\varepsilon \sim N(0, I)$
- C) $z = \sigma^2 + \varepsilon$
- D) $z = \mu - \varepsilon$

Answer: B.

Q3. A 28 × 28 image has how many input pixel values?

- A) 28
- B) 56
- C) 392
- D) 784

Answer: D. 28 × 28 = 784.

Q4. An entangled latent representation is one where:

- A) each latent variable controls exactly one factor
- B) changing one latent variable may change several properties together
- C) the latent size equals the input size
- D) the KL term is zero

Answer: B.

Q5. In a disentangled latent space, if only z1 (hair) is changed:

- A) all properties of the image change equally
- B) the image becomes random noise
- C) mainly the hair changes and the other properties stay mostly unchanged
- D) the decoder stops working

Answer: C.

Q6. The main benefit of a disentangled latent space is that it is:

- A) interpretable and controllable
- B) smaller than the input
- C) always perfectly reconstructive
- D) free of KL loss

Answer: A.

Q7. For a standard VAE (KL weight 1), which statement is correct?

- A) It always gives a disentangled latent space
- B) It never uses a KL term
- C) It can't reconstruct images
- D) It does not guarantee disentanglement

Answer: D. The model may still accept mixed information across latents if that helps reconstruction.

Q8. β-VAE changes the standard VAE loss by:

- A) multiplying the reconstruction term by β
- B) multiplying the KL term by β
- C) removing the KL term
- D) adding a label y

Answer: B.

Q9. In β-VAE for disentanglement, β is chosen:

- A) $\beta < 0$
- B) $\beta = 0$
- C) $\beta > 1$
- D) $\beta = 1$ exactly

Answer: C.

Q10. The reconstruction loss encourages:

- A) the encoder distribution to equal the prior
- B) the decoder to reconstruct the input, so $\hat{x} \approx x$
- C) the latent size to shrink
- D) the labels to be one-hot

Answer: B.

Q11. The KL loss encourages:

- A) $q_\varphi(z \mid x)$ to stay close to the standard normal prior
- B) the decoder to ignore z
- C) σ^2 to become 0
- D) $\hat{x}$ to differ from x

Answer: A.

Q12. A larger β puts stronger pressure on the encoder to have:

- A) $\mu(x)$ far from 0
- B) $\mu(x) \approx 0$ and $\sigma^2(x) \approx 1$
- C) $\sigma^2(x) \approx 0$
- D) larger latent size

Answer: B.

Q13. If β is too large, the most likely side effect is that:

- A) reconstruction quality decreases
- B) the latent space becomes larger
- C) the KL term disappears
- D) the model needs labels

Answer: A. The model focuses too much on staying close to N(0, I) and may not store enough detail.

Q14. With $\sigma^2 = 1$, the KL for one latent dimension simplifies to:

- A) μ^2
- B) $\frac{1}{2}\mu^2$
- C) $2\mu^2$
- D) $\frac{1}{2}\sigma^2$

Answer: B.

Q15. For $\mu = [0.4, 0.2]$ and $\sigma^2 = 1$, D_{KL} equals:

- A) 0.05
- B) 0.20
- C) 0.40
- D) 0.10

Answer: D. ½ (0.16 + 0.04) = 0.10.

Q16. For $\mu = [0.8, 0.6]$ and $\sigma^2 = 1$, D_{KL} equals:

- A) 0.50
- B) 1.00
- C) 0.25
- D) 0.70

Answer: A. ½ (0.64 + 0.36) = 0.50.

Q17. For $\mu = [1.5, 0.5]$ and $\sigma^2 = 1$, D_{KL} equals:

- A) 1.00
- B) 1.25
- C) 2.50
- D) 0.75

Answer: B. ½ (2.25 + 0.25) = 1.25.

Q18. A larger KL value in the lecture's examples means the encoder distribution is:

- A) closer to the prior and less informative
- B) identical to the prior
- C) farther from the prior and using more latent capacity
- D) independent of x

Answer: C.

Q19. An image has 6 important factors but the latent size is only 3, and disentanglement is enforced. What typically happens?

- A) All 6 factors are captured perfectly
- B) Only a few dominant factors are captured and reconstruction loss increases
- C) The KL term increases
- D) The decoder receives extra information

Answer: B.

Q20. Which of these is **not** a proposed fix for the trade-off?

- A) Increase the latent size
- B) Tune β to a moderate value
- C) Use β annealing

- D) Remove the decoder

Answer: D.

Q21. β annealing means:

- A) starting with a large β and reducing it
- B) keeping β fixed at 1
- C) starting with a small β and increasing it gradually
- D) setting β to a random value each epoch

Answer: C. The model learns reconstruction first, then the KL term is activated.

Q22. Which statement about β-VAE is correct?

- A) It guarantees perfect disentanglement
- B) It encourages disentanglement through stronger regularization, without guaranteeing it
- C) It requires class labels
- D) It reduces the KL term to zero

Answer: B.

Lecture 27: conditional VAE

Q23. A cVAE adds which extra input to a VAE?

- A) a noise vector
- B) a second image
- C) a condition y such as a class label or attribute
- D) a larger latent size

Answer: C.

Q24. A cVAE learns the mapping:

- A) $x \rightarrow z \rightarrow \hat{x}$
- B) $(x, y) \rightarrow z \rightarrow \hat{x}$
- C) $y \rightarrow x$ only
- D) $z \rightarrow y$

Answer: B.

Q25. The cVAE encoder models:

- A) $q_\varphi(z \mid x)$
- B) $p_\theta(x \mid z)$
- C) $q_\varphi(z \mid x, y)$
- D) $p(y)$

Answer: C.

Q26. The cVAE decoder models:

- A) $p_\theta(x \mid z, y)$
- B) $p_\theta(x \mid z)$
- C) $q_\varphi(z \mid x)$
- D) $p(z)$

Answer: A.

Q27. The main purpose of a cVAE is:

- A) to avoid using a KL term
- B) to compress images more
- C) to remove the decoder
- D) controlled generation

Answer: D.

Q28. Which of these is an example of a condition y?

- A) a class label such as the digit 5
- B) the reconstruction loss
- C) the learning rate
- D) the KL value

Answer: A.

Q29. In a cVAE the latent vector is sampled as:

- A) $z = \mu(x) + \sigma(x) \odot \varepsilon$
- B) $z = \mu(x, y) + \sigma(x, y) \odot \varepsilon$
- C) $z = y$
- D) $z = x \oplus y$

Answer: B.

Q30. Which is the cVAE loss on the slides?

- A) $-E[\log p_\theta(x \mid z)] + D_{KL}(q_\varphi(z \mid x) \,\|\, p(z))$
- B) $-E[\log p_\theta(x \mid z, y)] + D_{KL}(q_\varphi(z \mid x, y) \,\|\, p(z))$
- C) $-E[\log p_\theta(y \mid z)] + D_{KL}(p(z) \,\|\, q_\varphi(z \mid y))$
- D) $\beta \cdot D_{KL}$ only

Answer: B. y appears in the encoder and decoder terms, and the prior p(z) is unchanged.

Q31. Compared with the standard VAE loss, the cVAE loss differs in that:

- A) the KL term is removed
- B) the prior p(z) is replaced by p(y)
- C) the encoder and decoder terms are also conditioned on y
- D) β is added

Answer: C.

Q32. For x = [5.1, 3.5, 1.4, 0.2] and y = [1, 0], the cVAE encoder input is:

- A) [5.1, 3.5, 1.4, 0.2]
- B) [5.1, 3.5, 1.4, 0.2, 1, 0]
- C) [1, 0, 5.1, 3.5]
- D) [6.1, 3.5, 1.4, 0.2]

Answer: B. The label is concatenated to x.

Q33. For z = [0.917, 0.153] and y = [1, 0], the cVAE decoder input is:

- A) [0.917, 0.153]
- B) [0.917, 0.153, 1, 0]
- C) [1.917, 0.153]
- D) [1, 0, 0.917]

Answer: B.

Q34. On the slide the encoder hidden activation changes from [1.07, 1.93, 0.87] to [1.27, 1.83, 1.17] when the label is added. What does this show?

- A) The label has no effect
- B) The label contributes additional weighted information to the hidden layer
- C) The KL term has increased
- D) The decoder was removed

Answer: B.

Q35. The slide's cVAE reconstructed output is:

- A) [0.103, −0.028, 0.206, 0.243]
- B) [1.27, 1.83, 1.17]
- C) [0.147, −0.047, 0.280, 0.337]
- D) [0.917, 0.153, 1, 0]

Answer: C. Option A is the standard VAE output.

Q36. A standard VAE's generation is called not explicitly controlled because:

- A) it is trained only with the input data x, without a condition
- B) it has no decoder
- C) it has no latent space
- D) it uses labels

Answer: A.

Q37. The symbol ⊕ in the cVAE diagram means:

- A) addition of probabilities
- B) element-wise product
- C) subtraction
- D) concatenation of the label with the input or latent vector

Answer: D.

Q38. An MNIST cVAE uses 784 pixels and a one-hot label over 10 classes. The encoder input size is:

- A) 784
- B) 794
- C) 10
- D) 7,840

Answer: B.

Q39. To generate a specific digit with a trained cVAE you would:

- A) retrain it for each digit
- B) average two images
- C) sample z, attach the label for that digit, and decode [z, y]

- D) set the KL term to zero

Answer: C. (Standard practice; the slides state the goal as controlled generation.)

Q40. The standard VAE learns the mapping:

- A) $(x, y) \rightarrow z \rightarrow \hat{x}$
- B) $x \rightarrow z \rightarrow \hat{x}$
- C) $y \rightarrow z \rightarrow x$
- D) $z \rightarrow x \rightarrow y$

Answer: B.

Lecture 28: latent space interpolation

Q41. Latent space interpolation means:

- A) averaging the pixel values of two images
- B) training two VAEs
- C) increasing the latent size
- D) moving smoothly between two points in the learned latent space

Answer: D.

Q42. The interpolation formula is:

- A) $z_\alpha = (1 - \alpha)\, z_A + \alpha\, z_B$
- B) $z_\alpha = \alpha\, z_A + \alpha\, z_B$
- C) $z_\alpha = z_A - z_B$
- D) $z_\alpha = z_A \times z_B$

Answer: A.

Q43. When $\alpha = 0$, z_α equals:

- A) z_B
- B) the midpoint
- C) z_A
- D) 0

Answer: C.

Q44. When $\alpha = 1$, z_α equals:

- A) z_A
- B) z_B
- C) the midpoint
- D) 1

Answer: B.

Q45. With z_A = [0.764, 0.318] and z_B = [−0.200, 1.100], z at α = 0.25 is:

- A) [0.282, 0.709]
- B) [0.041, 0.9045]
- C) [0.523, 0.514]
- D) [0.764, 0.318]

Answer: C.

Q46. For the same points, z at α = 0.5 is:

- A) [0.282, 0.709]
- B) [0.523, 0.514]
- C) [0.041, 0.9045]
- D) [0.564, 0.709]

Answer: A. It is the average of the two points.

Q47. For the same points, z at α = 0.75 is:

- A) [0.523, 0.514]
- B) [0.041, 0.9045]
- C) [−0.200, 1.100]
- D) [0.282, 0.709]

Answer: B.

Q48. After interpolation, each interpolated latent vector is passed to:

- A) the encoder
- B) the KL term
- C) the decoder
- D) the classifier

Answer: C.

Q49. The images generated for increasing α are expected to go:

- A) digit 3 → 3-like → mixed 3/8 → 8-like → digit 8
- B) digit 8 → digit 3 → noise
- C) digit 3 → digit 3 → digit 3
- D) random digits

Answer: A.

Q50. In the lecture, interpolation takes place:

- A) in pixel space
- B) in the label space
- C) in the loss function
- D) in the latent space

Answer: D.

Q51. In the lecture's example, z_A represents:

- A) image A, the digit 3
- B) image B, the digit 8
- C) the average of the two
- D) the noise ε

Answer: A.

Q52. The latent space in the interpolation example has how many dimensions?

- A) 2
- B) 4
- C) 784
- D) 1

Answer: A. z = [z1, z2].

Q53. The interpolation parameter α controls:

- A) the number of latent dimensions
- B) how far we move from z_A toward z_B
- C) the learning rate
- D) the KL weight

Answer: B.

Q54. In interpolation, the decoder's job is to:

- A) compute the interpolation parameter
- B) sample ε
- C) classify the digit
- D) map each latent vector back to an image

Answer: D.

Mixed questions

Q55. You need to generate a specific digit class on demand. Which model fits best?

- A) a standard VAE
- B) a conditional VAE
- C) a β-VAE
- D) interpolation

Answer: B.

Q56. Which model is designed to make individual latent dimensions interpretable?

- A) cVAE
- B) a plain autoencoder
- C) β-VAE
- D) latent space interpolation

Answer: C.

Q57. Which change most directly forces the encoder distribution toward the prior?

- A) a larger β
- B) a smaller β
- C) adding a label y
- D) interpolating with $\alpha = 0.5$

Answer: A.

Q58. In training a β-VAE with $\beta > 1$, an encoder with $\mu = [1.5, 0.5]$ would feel pressure to:

- A) increase its KL
- B) increase its latent size

- C) move μ toward 0, lowering the KL
- D) ignore the decoder

Answer: C.

Q59. Which pairing is correct?

- A) cVAE: weights the KL term by β
- B) β-VAE: adds a condition y
- C) interpolation: a weighted average of two latent vectors
- D) disentanglement: guaranteed by the standard VAE

Answer: C.

Q60. A student creates an intermediate image by averaging the pixel values of a 3 and an 8. This is:

- A) latent space interpolation
- B) conditional generation
- C) β annealing
- D) not latent space interpolation, because it happens in pixel space

Answer: D.

Part 5: select-all-that-apply and true/false

MSQ (select all that apply)

M1. Which statements about β-VAE are true?

- A) It uses $\beta > 1$
- B) It weights the KL term
- C) It guarantees perfect disentanglement
- D) A very large β can reduce reconstruction quality

Answer: A, B, D. C is false: it encourages disentanglement but doesn't guarantee it.

M2. Which are inputs to the cVAE encoder?

- A) x
- B) y
- C) the noise ε

- D) the latent z

Answer: A, B. ε is used when sampling z, and z is the encoder's output.

M3. When the image has more factors than the latent space has variables, which are valid fixes?

- A) Increase the latent size
- B) Choose a moderate β
- C) Use β annealing
- D) Remove the KL term entirely

Answer: A, B, C.

M4. Which statements about $z_\alpha = (1 - \alpha)\, z_A + \alpha\, z_B$ are true?

- A) $\alpha = 0$ gives z_A
- B) $\alpha = 1$ gives z_B
- C) The two weights always sum to 1
- D) α must be larger than 1

Answer: A, B, C.

M5. Which describe an entangled latent representation?

- A) One latent variable holds a mix of factors
- B) Changing one latent variable changes several properties
- C) It is easy to interpret and control
- D) A standard VAE with KL weight 1 can still end up with it

Answer: A, B, D.

M6. The KL value goes up when:

- A) μ moves away from 0
- B) σ^2 moves away from 1
- C) $\mu = 0$ and $\sigma^2 = 1$
- D) the distribution moves farther from the prior

Answer: A, B, D. C is the case where KL is exactly 0.

M7. Which are true of the cVAE decoder?

- A) It takes z

- B) It takes y
- C) It models $p_\theta(x \mid z, y)$
- D) It ignores z

Answer: A, B, C.

M8. Which are examples of latent space interpolation from the lecture?

- A) A smooth transition from the digit 3 to the digit 8
- B) A morph from one 3D chair to another
- C) Averaging the pixel values of two images
- D) Decoding intermediate latent vectors

Answer: A, B, D.

M9. Which of these values from the slides are correct?

- A) $D_{KL} = 0.10$ for $\mu = [0.4, 0.2]$, $\sigma^2 = 1$
- B) The cVAE encoder input in the example has 6 values
- C) $z_{0.5} = [0.282, 0.709]$
- D) A 28 × 28 image has 784 pixel values

Answer: A, B, C, D.

M10. Increasing β:

- A) raises the weight of the KL term
- B) tends to lower the KL value
- C) tends to improve reconstruction
- D) pushes $q_\varphi(z \mid x)$ toward $N(0, I)$

Answer: A, B, D. C is the reverse of what happens.

True or false

1. A standard VAE guarantees a disentangled latent space. **False.**
2. In β-VAE, β is greater than 1. **True.**
3. With $\sigma^2 = 1$, the KL for one dimension simplifies to $\frac{1}{2}\mu^2$. **True.**
4. A larger KL value means the distribution is closer to the prior. **False.** It means farther.
5. The cVAE decoder uses both z and y. **True.**

6. On the slides, the cVAE replaces the prior p(z) with a label-dependent prior. **False.** The prior stays p(z).
7. z at α = 0.5 is the average of z_A and z_B. **True.**
8. Latent space interpolation is done on the pixel values of the two images. **False.** It is done on latent vectors.
9. β annealing starts with a large β and reduces it. **False.** It starts small and increases.
10. Increasing the latent size can lower reconstruction loss when there are more factors than latent variables. **True.**
11. In an entangled representation each latent controls exactly one factor. **False.**
12. Doubling μ doubles the KL (with $\sigma^2 = 1$). **False.** It quadruples it.
13. When ε = 0, z = μ. **True.**
14. In a cVAE the label is concatenated to the input and to z. **True.**
15. α > 1 gives a point between z_A and z_B. **False.** It extrapolates beyond z_B.

Part 6: short and long answer questions

Short answers

1. Define entangled and disentangled latent representations. Entangled: each latent variable holds a mix of several factors, so changing one changes multiple properties. Disentangled: each latent dimension mainly captures one meaningful factor, so changing one changes mainly one property. A disentangled space is interpretable and controllable.

2. Why doesn't a standard VAE guarantee disentanglement? The KL term is present but has weight 1. The model may still accept mixed information spread over many latents if that helps reconstruction, so the latents can stay entangled.

3. What is the role of β in β-VAE? β multiplies the KL term (β > 1). It puts stronger pressure on $q_\varphi(z \mid x)$ to stay close to N(0, I), pushing μ toward 0 and σ^2 toward 1, which restricts the model from storing mixed information freely and encourages disentanglement.

4. What happens when β is too large, and what can you do? The model focuses too much on staying close to the prior and doesn't store enough detail, so reconstruction quality drops. Remedies: increase the latent size, choose a moderate β, or use β annealing.

5. Write the cVAE loss and say how it differs from the VAE loss. `L_cVAE(x) = -E_q(z|x,y)[log pθ(x | z, y)] + D_KL(qφ(z | x, y) || p(z))`. The VAE loss has the same form without y. In the cVAE both the encoder distribution and the decoder likelihood are conditioned on the label.

6. What does y do in the cVAE encoder and decoder? In the encoder it is concatenated with x, so qφ(z | x, y) depends on both. In the decoder it is concatenated with z, so pθ(x | z, y) generates data according to the given condition.

7. Define latent space interpolation and give the formula. Moving smoothly between two latent points. Encode two inputs to z_A and z_B, then compute `z_α = (1 − α) z_A + α z_B` for values of α between 0 and 1, and decode each vector.

8. Why are interpolated vectors passed to the decoder? The interpolated vectors are points in latent space, not images. The decoder maps each one back to the data space, giving the sequence of intermediate images.

9. Explain β annealing. Instead of using a high β from the start, begin with a small value (for example 1 and then 2 and then 4, or starting from 0) so the model first learns good reconstruction, then increase β gradually so the KL term becomes active and improves the latent structure.

10. Give a case where the KL increases while the reconstruction improves. When the image has many factors but the latent space is small and disentanglement isn't enforced (entanglement case): the latents carry more information, so reconstruction loss is lower, but the encoder distribution moves farther from the prior, so the KL increases.

Long answers (model-answer outlines)

L1. Explain entangled and disentangled latent spaces with an example, and how β-VAE encourages disentanglement.

- Define both, with the digit example (thickness, slant, rotation, position) or the face example (hair, lighting, skin tone, smile, pose).
- Show the entangled architecture: every hidden unit combines all 784 pixels, so every latent combines all factors.
- Explain the face example: changing z1 changes mainly the hair if disentangled.
- Give the standard VAE and β-VAE losses, and explain what each term does.
- Say why β > 1 helps: stronger KL pressure toward N(0, I), limiting freely stored mixed information.
- Close with the caveats: no guarantee, and a too-large β hurts reconstruction.

L2. Compute and interpret the KL for μ = [0.4, 0.2], [0.8, 0.6] and [1.5, 0.5] with $\sigma^2 = 1$.

- Write the general formula and simplify for $\sigma^2 = 1$.
- Compute 0.10, 0.50 and 1.25 with the substitutions shown.
- Interpret: a small KL means close to the prior and more regularized, like the disentangled example. A large KL means farther from the prior and more latent capacity used, like the entangled examples.

L3. Discuss the trade-off when an image has 6 factors but the latent size is 3.

- Case 1, entanglement: mixed factors, lower reconstruction loss, higher KL, and β too low to enforce disentanglement.
- Case 2, disentanglement enforced: only a few dominant factors captured, reconstruction loss increases, KL decreases with $\beta > 1$.
- Fixes: increase the latent size (3 to 6), tune β to a moderate value, use β annealing, with a sentence on why each helps.

L4. Explain the conditional VAE: architecture, loss and the numerical example.

- Motivation: a standard VAE's generation isn't explicitly controlled.
- Architecture: encoder $q_\varphi(z \mid x, y)$, decoder $p_\theta(x \mid z, y)$, concatenation, and the sampling formula.
- The loss with y added.
- Numbers: [x, y] = [5.1, 3.5, 1.4, 0.2, 1, 0]; hidden activation [1.07, 1.93, 0.87] to [1.27, 1.83, 1.17]; decoder input [0.917, 0.153, 1, 0]; outputs [0.103, −0.028, 0.206, 0.243] versus [0.147, −0.047, 0.280, 0.337].
- Conclusion: the label contributes additional weighted information and changes the output.

L5. Explain latent space interpolation with a worked example.

- The idea: two images, two latent points, move in steps, decode.
- z_A = [0.764, 0.318] (digit 3), z_B = [−0.200, 1.100] (digit 8).
- The formula, then the $\alpha = 0.25$ calculation in full: [0.523, 0.514].
- The table for α = 0, 0.25, 0.5, 0.75, 1.
- The decoded sequence from 3 to 8 and why it is smooth.

L6. Compare the VAE, β-VAE and cVAE.

- Use the comparison table from Part 1: what each learns, what changes, its goal and its weakness.

- Add one line saying they can be combined.

Part 7: last-minute checklist and revision plan

Things to know cold

- The reparameterization formula and what ⊙ means.
- The VAE, β-VAE and cVAE losses, and which part changes in each.
- KL for $\sigma^2 = 1$ is $\frac{1}{2} \Sigma \mu^2$. Know the lecture results: **0.10, 0.50, 1.25**.
- $\beta > 1$: more KL weight, closer to the prior, worse reconstruction if too large.
- The three fixes: increase latent size, moderate β, β annealing.
- cVAE: [x, y] into the encoder, [z, y] into the decoder, and the lecture vectors.
- Interpolation: `z_α = (1 − α) z_A + α z_B`, and the five-row table.
- Interpolation is in latent space, never pixel space.

Revision plan

These are sessions, not fixed dates, so fit them around your other exams.

1. **Session 1 (about 60 to 90 minutes).** Read Part 1 and Part 2 once. Close the doc and write the formula sheet from memory, then check it. Do numericals N1 to N13.
2. **Session 2.** Do numericals N14 to N32 and MCQs Q1 to Q40. For every miss, find the slide idea behind it and write one line.
3. **Session 3 (the day before).** Do MCQs Q41 to Q60, the MSQs and true/false. Redo any numericals you missed. Read the common mix-ups in Part 1.
4. **Exam morning.** Only the formula sheet and the numbers listed above.

If you're short on time, do this order: formulas, the lecture numbers (KL values, interpolation table, cVAE vectors), then the β trade-off, then as many MCQs as you can.

Want more practice on any part? Tell me which lecture and whether you want MCQs, numericals or long answers, and I'll add them to this doc.